

LEAST SQUARES (LS).

Consider the system of Linear Equations

$$\begin{matrix} m = \# \text{ of observations} \end{matrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = A \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \begin{matrix} \leftarrow \# \text{ of unknowns} \\ \# \text{ of inputs} \end{matrix}$$

$m \times n$

$$\begin{matrix} m \times 1 \end{matrix} \overline{y} = \begin{matrix} m \times n \end{matrix} A \begin{matrix} n \times 1 \end{matrix} \overline{x}$$

if $m = n$ & A is invertible

then, the solution is,

$$\overline{x} = A^{-1} \overline{y}.$$

However, if $m > n$,
then there are more equations
than unknowns.

$\# \text{ outputs} > \# \text{ inputs}$

$\Rightarrow$ this is termed as an
OVER DETERMINED system.

What then is $\overline{x}$?

How to determine $\overline{x}$?

Can you determine $\bar{x}$?

$$\bar{y} = \begin{bmatrix} \bar{a}_1 & \bar{a}_2 & \dots & \bar{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

n columns

$$= \underbrace{\bar{a}_1 x_1 + \bar{a}_2 x_2 + \dots + \bar{a}_n x_n}_{\text{linear combination of } \bar{a}_1, \bar{a}_2, \dots, \bar{a}_n. \sim C(A)}$$

$\Rightarrow A\bar{x}$ spans n dimensional subspace

$C(A)$ = column space of A .

However, $\bar{y} \in m$ dimensional space!

Solution exists only if

$$\bar{y} \in C(A).$$

else if $\bar{y} \notin C(A)$, then solution does NOT exist.

What is the best that you can do?

$$\bar{y} \sim A\bar{x} \quad \begin{array}{l} \nearrow \text{Determine an} \\ \nearrow \text{approximate soln.} \end{array}$$

Best possible approximation!

$$\bar{y} - A\bar{x} = \bar{e}$$

$\hookrightarrow$ approximation error.

$$\text{minimize } \|\text{approx. error}\| \equiv \min. \|\bar{e}\|$$

$$\equiv \min. \|\bar{e}\|^2$$

$$\equiv \min. \bar{e}^T \cdot \bar{e}$$

$$\| \bar{y} - H\bar{x} \|^2$$

'Least Squares' (LS)

$$\begin{aligned} \|\bar{e}\|^2 &= \bar{e}^T \cdot \bar{e} = (\bar{y} - H\bar{x})^T (\bar{y} - H\bar{x}) \\ &= (\bar{y}^T - \bar{x}^T H^T) (\bar{y} - H\bar{x}) \\ &= \bar{y}^T \bar{y} - \bar{x}^T H^T \bar{y} - \bar{y}^T H \bar{x} \\ &\quad + \bar{x}^T H^T H \bar{x} \end{aligned}$$

$$f(\bar{x}) \leftarrow = \left\{ \bar{y}^T \bar{y} - 2 \bar{x}^T H^T \bar{y} + \bar{x}^T H^T H \bar{x} \right\}$$

objective function

find $\bar{x}$ which minimizes $f(\bar{x})$

$$\min. f(\bar{x})$$



KKT framework.

$$\nabla f(\bar{x}) = 0$$

$$\nabla f = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \\ \vdots \\ \partial f / \partial x_n \end{bmatrix} = 0.$$

$$\Rightarrow \frac{\partial f}{\partial x_i} = 0.$$

Derivative w.r.t. every component

of $\bar{x}$ is 0.

Properties of ∇ :

Linear $\Rightarrow \nabla (\bar{c}^T \bar{x}) = \nabla (\bar{x}^T \bar{c})$
 $= \nabla (c_1 x_1 + c_2 x_2 + \dots + c_n x_n)$
 $= \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \bar{c}$

$\nabla (\bar{x}^T P \bar{x}) = (P + P^T) \bar{x}$
 $= 2P \bar{x} \quad (\text{if } \underbrace{P = P^T}_{P \text{ is symmetric}})$
Quadratic

$f(\bar{x}) = \underbrace{\bar{y}^T \bar{y}}_{\text{constant}} - 2 \underbrace{\bar{x}^T A^T \bar{y}}_{\text{Linear}} + \underbrace{\bar{x}^T A^T A \bar{x}}_{\text{Quadratic}}$

$\nabla f(\bar{x}) = 0 - 2 A^T \bar{y} + 2 A^T A \bar{x} = 0$

$(A^T A)^T = A^T A$

$A^T A$ is symmetric

$\Rightarrow A^T A \bar{x} = A^T \bar{y}$

$\Rightarrow \bar{x} = (A^T A)^{-1} A^T \bar{y}$

Least Squares solution.

if $A^T A$ is invertible
 True if A is full column rank
 $\Rightarrow \text{rank}(A) = n.$

$$\bar{x} = \underbrace{(A^T A)^{-1} A^T \bar{y}}_{\text{Least Squares soln. (LS)}}$$

daguer
 $\nearrow$

$$A^\dagger = (A^T A)^{-1} A^T = \text{Pseudo-inverse}$$

Because

$$\begin{aligned} & \underbrace{(A^T A)^{-1} A^T} \cdot A \\ &= (A^T A)^{-1} (A^T A) \\ &= \underline{I} \end{aligned}$$

acts like an inverse!

A is NOT invertible
 if $\underbrace{m > n.}$

i.e. A is NOT square.

$\left[\begin{array}{c} \\ \\ \end{array} \right] \leftarrow \begin{array}{l} \text{more rows} \\ \text{than columns} \end{array}$
 $\longrightarrow \text{tall matrix.}$

Note: $(A^T A)^{-1} A^T A = \underline{I}$

gives I only when multiplied
 on left $\Rightarrow$ Left inverse of A .

